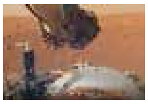
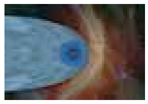


InSight flexes its arms
NASA's InSight's six-feet long robotic arm will be use to pick up and drop scientific equipment onto Mars' surface. **https://dgit.in/jan19-71**



Voyager 2 in the beyond
NASA's sVoyager 2 probe which has been in space for over four decades has now entered interstellar space. **https://dgit.in/jan19-72**

How VR treatment alleviates actual pain

Virtual reality has made strides through gaming and entertainment, however, newer applications in the medical field are blooming that enhance the pain treatment processes.

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Virtual reality is a highly coveted and experimented upon computer technology that creates an artificial 3-D simulated environment. Usually featuring a head-mounted display and a pair of goggles attached to a computer or smartphone, the applications of VR are pushing past their original intended purpose. Physicians and doctors have imparted VR treatment or therapy as a capable option to diminish actual pain and stress in patients in many parts of the world. Patients are placed into a 'virtual world' immersing their visual and auditory senses, thereby distracting them from the pain they experience. The exploration into this field of VR application is in its preliminary stages, however, a fair amount of studies and experiments seem to validate the authenticity of this unorthodox method.

HOW IT WORKS

Pain is a highly complex experience with the physical permutations being at the forefront but also comprising of cognitive, behavioural and psychological components. VR application in the medical field focuses on many varying degrees of pain caused by procedures such as vaccinations, intravenous injections, lacerations, dressing changes for burn victims and stitches. It is also used to treat chronic pain, psychosomatic pain, phantom pain (occurring when limbs or arms are amputated) and other such psychological pain conditions.

VR treatment attempts to improve 'mirror therapy' used to alleviate the pain of patients who have undergone amputation.

VR therapy or treatment builds upon previously utilised techniques in this field like distractions and 'mirror therapy'. 80 per cent amputees suffer from phantom limb pain. To treat this, mirror therapy was used which traditionally refers to a patient looking at the reflection of the unaffected limb creating an illusion that both their limbs are healthy and movements take place without pain. VR builds upon this by helping them visualise a lot more than a mere reflection. It would eliminate the need of patients having to rely on their imagination or a box. The amputated patients would put on VR glasses and have sensors attached to the nerve endings of their affected limb, which would, in turn, send signals to the brain about the movement it wanted to accomplish. Patients would